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Prompt-NO formation in methane/oxygen/nitrogen flames seeded with oxygenated volatile organic compounds: Methyl ethyl ketone or ethyl acetate

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Abstract

In the present work, CH and NO profiles are determined using laser-induced fluorescence (LIF) measurements in eight low-pressure laminar flames of $CH_4/O_2/N_2$ containing various amounts of methyl ethyl ketone or ethyl acetate with respect to the equivalence ratio. Relative CH LIF signals are calibrated using cavity ring-down spectroscopy (CRDS), while NO LIF calibration is performed in the burned gases of NO seeded flames. Temperature measurements are obtained using a coated Pt/Rh thermocouple and serve as temperature profiles input for Chemkin modeling. Volatile organic compound (VOC) submechanisms, previously validated upon major and intermediate species profiles measured using sampling techniques, are incorporated into the GDF-Kin 3.0_NCN for the CH_4 oxidation. This mechanism is adjusted and validated to predict the effect of VOCs seeding on CH and NO formation. It takes into account not only the CH and NO profiles but those previously measured. When methane is replaced by either MEK or EA, the NO mole fraction in the burned gases and the CH peak value are found to jointly decrease, indicating that NO is mainly formed according to the prompt-NO mechanism. According to the kinetic analysis, the VOC impact on NO formation is demonstrated. It is shown that CH radical formation, through the C1 sequence, mainly involves the CH_3 radical, which is formed with the same propensity by one molecule of methane, MEK, or EA. Due to the methane replacement by VOC while the equivalence ratio is maintained constant, we found that the CH peak value decrease follows the total fuel volumetric flow rate decrease, yielding, an NO decrease in the burned gases.
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Keywords: Prompt-NO; Oxygenated VOC; CH; NCN; LIF; Chemical kinetic mechanism

1. Introduction

Nowdays, tremendous efforts are made to prevent pollutant emissions in the atmosphere. Volatile organic compounds (VOCs) are known to be precursors of ozone formation in the troposphere under UV radiations in the presence of NO_x. VOCs

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are mainly formed from either transportation emissions or solvent industries. Different ways of degradation of VOCs are generally proposed in order to prevent their release in air: catalytic oxidation or thermal degradation. Thermal degradation requires a relatively high level of temperature (700–1700 K), which means fuel consumption and NO_x formation. This process presents the main advantage that it works continuously and is able to treat a large quantity of compounds without restrictions about their nature, unlike the catalytic processes. Two different thermal processes are proposed for industrial applications; regenerative [1] or recuperative [2]. In order to better overcome NO_x formation, it is necessary to have a better knowledge of the VOCs' kinetic chemistry.

Only a few studies have been performed on the influence and/or effect of oxygenated compounds on NO_x formation. In a well-stirred reactor at a temperature lower than 1450 K, it has been shown that in the presence of oxygenated compounds (involving different kinds of C–O bonds), NO reduction by urea can be obtained not only at a lower temperature but also within a wider temperature range [3]. The efficiency of the NO reduction was shown to depend on the additive but was found to be similar for methyl acetate and propylaldehyde (both C3 species, respectively CH_3COOCH_3 and C_2H_5CHO). In a small-scale recuperative incinerator, Salvador et al. [2] conducted a parametric study to improve the operating conditions of industrial devices. The VOCs are added to the heated air flux before entering into a burner supplied with natural gas. Different parameters were considered. Only ethyl acetate was examined in this study. They showed that NO_x emissions decrease when the ethyl acetate flow rate in air increases while the natural gas supply to the burner decreases.

The present work aims to propose a detailed kinetic model for the oxygenated VOCs' oxidation in the presence of N-species in order to predict NO formation in low-pressure methane/O_2/N_2 flames seeded with VOCs. Two oxygenated compounds, containing the same number of C-atoms and H-atoms, representative of ketones and esters were chosen: methyl ethyl ketone (MEK), which is commonly used in paint and adhesive manufacture and application, and ethyl acetate (EA), which is commonly used as a solvent, notably in some segments of the printing industry. The detailed kinetic scheme is based on two kinds of flame structure studies performed in our laboratory. The first one was devoted to the fulfillment of a kinetic mechanism describing methyl ethyl ketone ($CH_3COC_2H_5$) and ethyl acetate ($CH_3COOC_2H_5$) oxidation [4,5]. Experiments were conducted in low-pressure laminar flames seeded with either MEK or EA using probe sampling techniques [6,7]. The second one, dedicated to NO formation in natural gas flames, made it possible to propose a new detailed kinetic mechanism, GDF-Kin 3.0_NCN [8]. Experiments were performed in natural gas/air flames and focused on the quantitative measurements, using laser induced fluorescence (LIF), of two major species involved in the prompt-NO mechanism: CH and NO [9].

In this work, several methane/O_2/N_2 flames seeded with either MEK or EA have been stabilized and analyzed in terms of NO formation by measuring NO and CH radicals by laser diagnostics. During this work, flames have been analyzed at a relatively lean equivalence ratio with respect to fuel consumption. Meanwhile, NO formation will be limited. The initial detailed kinetic scheme has been modified to better predict experimental CH and NO profiles. Modifications were performed with respect to species profiles previously measured by GC techniques. It will be shown that oxidation in the presence of oxygenated VOCs leads to NO diminution.

2. Experimental setup

Experiments were performed in different laminar, premixed flames stabilized at 5.3 kPa (40 Torr). The homemade 63-mm-diameter burner (the plate is equally perforated with 800-μm-diameter holes) is the one previously used for studies of flames containing VOCs [6,7]. The inner side is filled with small balls to better stabilize the flame. The burner is vertically mobile in a stainless enclosure. The setup (burner, enclosure, gas supply lines) was heated at 353 K. The pressure was controlled via a 0–13.3 kPa pressure gauge. The facility is equipped with four optical accesses for the laser diagnostics.

The mixture composition was controlled via mass flow meters. The liquid VOCs are preheated at 353 K in an enclosed stainless device in order to work with the vapor phase. Several flames have been studied. In a first series where MEK has been added to the mixture, the mixture equivalence ratio is close to 0.78 and the N_2/O_2 ratio is equal to 3.26. In a second series where EA has been added to the mixture, the equivalence ratio is close to 0.85 and the N_2/O_2 ratio is equal to 3.10. The N_2/O_2 ratio value varies only by 5% between the two series, which does not affect the temperature profiles. A problem in the mass flow controller calibration procedure was highlighted during the data process, explaining the difference of the equivalence ratio between the MEK and the EA seeded flames. In order to have a standard flame for modeling, the study was completed with a CH_4/O_2/N_2 flame, hereafter called the reference flame. Premixed gas compositions are given in Table 1.

Table 1
Flames mixture compositions

Flame	VOC	ϕ	N_2/O_2	Q	Q_{CH_4}	Q_{VOC}	Q_{O_2}	Q_{N_2}	Q_{fuel}	FLRT
Reference		0.94	3.48	269	25.9	0.0	54.8	190.4	25.90	2.96×10^{-3}
Ref/MEK		0.78	3.26	246	20.8	0.0	53.3	173.8	20.80	2.73×10^{-3}
1-MEK	MEK	0.80	3.26	246	18.5	1.1	53.6	174.9	19.55	2.76×10^{-3}
2-MEK	MEK	0.78	3.26	246	12.1	3.4	54.5	177.8	15.40	2.82×10^{-3}
3-MEK	MEK	0.76	3.26	245	5.5	5.6	55.4	180.8	11.10	2.89×10^{-3}
Ref/EA		0.85	3.10	246	23.3	0.0	54.8	169.9	23.29	2.72×10^{-3}
1-EA	EA	0.87	3.10	247	18.8	2.1	55.7	172.6	20.97	2.79×10^{-3}
2-EA	EA	0.84	3.10	247	11.8	4.8	56.7	175.9	16.62	2.87×10^{-3}

Note. Q represents the standard volumetric flow rate (L/h), ϕ is the equivalence ratio, FLRT is the mass flux ($g/cm^2/s$), and Q_{fuel} represents the total volumetric flow rate of fuel in the flame.

3. Temperature profiles

The temperature profiles, corrected for radiation, were determined using a 100-μm-diameter wire coated Pt-6%Rh/Pt-30%Rh thermocouple. The thermocouple was placed parallel to the burner surface. More details about the method and the radiation losses correction were given previously [6,10]. For example, at 20 mm above the burner in the reference flame, the temperature was determined to be 1680 K. Taking into account the radiation losses, the temperature was corrected to be 1890 K. The global accuracy of the temperature measurements is estimated to be about 5% along the temperature profile. Repeatability of the temperature measurements under each set of flame conditions was better than ±5 K.

4. Laser diagnostics

The laser system consists of a frequency-doubled Quantel Nd:YAG laser pumping a dye laser. Wavelengths around 317 nm for the CH radical excitation were provided by doubling the fundamental dye radiation (obtained with a mixture of dyes: Rh640 and DCM) pumped with 532 nm. For the NO excitation, wavelengths around 225 nm were obtained by mixing the frequency-doubled output of the dye laser (constituted by a mixture of Rh590 and Rh610) with the residual infrared radiation of the YAG laser. The resulting FWHM linewidth was around 1 cm^{-1}. Dye laser wavelengths were systematically calibrated with the LIF-excitation spectra using the LIFBASE software [11].

Species (CH and NO) profiles were locally measured by LIF. The laser beam was focused before entering into the burner enclosure (focal length 500 mm). The laser energy fluctuations were monitored by a postflame photodiode. The LIF signal was collected at $f/4$ by a two-lens system and focused on the entrance slit of a 0.25-m monochromator, providing a high degree of rejection of background radiation. The entrance slit was parallel to the laser axis, and the output slit was adjustable to provide a bandpass adapted to the fluorescence band under investigation. The fluorescence spectra were collected with a 1.5-ns rise-time Philips XP2020Q photomultiplier tube. The fluorescence and laser intensity signals were simultaneously resolved and averaged over 500 pulses and stored by a digital scope (LECROY 9354A, 8-bit, 500-MHz bandwidth, 1-GS/s sampling rate). LIF signals were measured at their temporal peak to minimize the effect of quantum yield variation along the flame on $N(J'')$ determination. The LIF signal being proportional to the rotational population $N(J'')$, conversion into total density number has been completed by dividing the signal by the Boltzmann fraction $f_B(J'', T)$, easily calculated using the experimental temperature data. Comparison with modeling data being based on the species mole fraction, this last one is derived from the total density and the ideal gas law. The specific conditions of the relative LIF measurements are given below.

The relative species profiles in each flame were measured by LIF. Thus, the peak LIF signals obtained in the different flames have been linked to each other by comparing their LIF intensity to that obtained in the reference flame (see Table 1), for which a direct calibration has been undertaken. The "linking" procedure has made it necessary to change the flame conditions between flame (i) and the reference flame and to compare the peak LIF signals. That procedure has been repeated several times to limit the uncertainty. The standard deviation was less than 3%. The calibration of the LIF signals obtained in the reference flame was performed in the zone where the species concentration is maximal. So it was performed at the peak value for the CH calibration and at 20 mm above the burner for the NO calibration.

4.1. CH measurements

CH LIF measurements were performed using the predissociative $C^2\Sigma^+ \leftarrow X^2\Pi(0,0)$ state near 315 nm, according to an excitation/detection scheme previously used in the lab [9,12]. The doublet $P_1(10)/P_2(10)$ near 317.3 nm was selected for the CH excitation because it is characterized by a weak variation of the ratio $f_B(J'',T)/T$ in the flame temperature range. CH measurements at low pressure are weakly affected by quenching variation because of the predissociative feature of the excited state. In order to reject all the interferences due to the flame emission, the fluorescence was collected in the entire Q-branch at 314.7 nm with a narrow bandwidth of 3.0 nm selected by the monochromator.

The calibration step was performed by cavity ring-down spectroscopy (CRDS) measurements in the reference flame. The laser beam was shaped with a system of a 100-μm pinhole aperture and two 100-mm focal lenses in order to match approximately the TEM_{00} transverse mode of the CRD cavity. The 390-mm-length CRD cavity consisted of two high-reflectivity mirrors (radius curvature 250 mm, $R = 99.7\%$ at 317 nm). The CRD signal was recorded using the same PMT and scope, both as in LIF measurements, and processed in order to measure the decay time of the CRD cavity. The measurements were performed at the CH-peak by scanning the laser around the $P_1(8)$ and $P_2(8)$ lines (on resonance). The CRD lifetime was found to be 400 ns with the laser tuned off resonance and decreased to about 335 ns with the laser tuned on resonance (with the $P_1(8)$ line) at the CH peak location. From the measured lifetimes on and off resonance, one obtains the integrated absorptivity according to the procedure previously described [13]. The $P_1(8)$ and $P_2(8)$ lines have been preferred because they are separated one from the other, unlike the doublet $P_1(10)/P_2(10)$, for which a deconvolution is required to determine the integrated absorptivity. From the integrated absorptivity of the lines $P_1(8)$ and $P_2(8)$ and considering the absorption constant across the flame diameter (one-dimensional flame), the absolute number density of the excited rotational level ($J'' = 8$) was determined. Knowing the temperature at the measured location ($T = 1725 \pm 100$ K), the CH-peak mole fraction was determined as being 2.7 ± 0.2 ppm in the reference flame.

As previously pointed out [9,14], due to the fact that the CRDS is a line-of-sight absorption technique, inhomogeneities along the laser beam in the entire flame diameter have to be taken into account, since they lead to underestimation of the concentration on the burner axis. Luque et al. [14] have shown, using planar LIF, that corrections could reach 20%. Similarly, Pillier et al. [9] have found a correction factor of 25%. In this work, assuming the same correction factor (1.25), the previous CH-peak mole fraction was corrected to 3.4 ± 0.25 ppm in the reference flame.

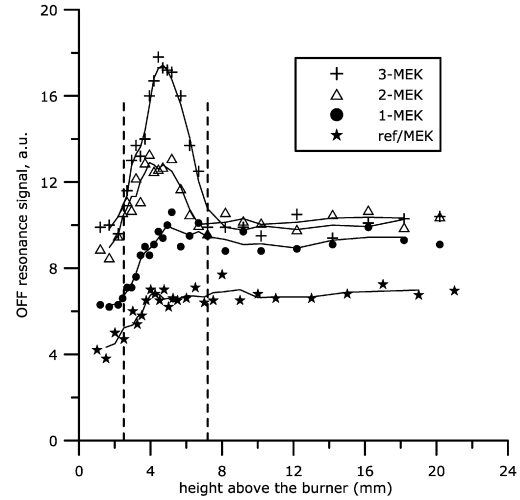

Fig. 1. Off-resonance profiles at 225.45 nm in the different flames containing MEK; see Table 1 for the full gas composition. Dashed lines show the reaction zone position based on the position of the peak LIF signal of the CH radical.

4.2. NO measurements

NO-LIF measurements in flames have been performed by excitation of the $A^2\Sigma^+ \leftarrow X^2\Pi(0\text{–}0)$ band ($Q_1(23)$ or $Q_2(27)$ lines) near 225.6 nm. The fluorescence collection was achieved using the $A^2\Sigma^+ \leftarrow X^2\Pi(0\text{–}2)$ band at 246 nm with a bandwidth of 8 nm selected by the monochromator. The excitation range is possibly disturbed by O_2 Schumann–Runge bands. Caution has been taken to prevent this background signal by recording the off resonance signals [15]. In the presence of MEK in the premixed gases, the off resonance (at 225.45 nm) profiles show an increase around the flame front zone (Fig. 1). This effect was negligible in flames containing EA. The background signal in the presence of MEK may be attributed to acetone emission and/or fluorescence, which was shown to be present in the flame front of MEK-containing flames from previous gas chromatography (GC) measurements [6]. The background signal in the flame 3-MEK could reach 55% of the total signal collected at 246 nm. It was subtracted on the basis of on–off measurements. Finally, the relative NO profiles were determined from the difference between the on and the off resonance signals in each flame, i.e., the off-resonance-corrected LIF signal. The LIF signals measured at 20 mm above the burner were

then related to the one measured in the reference flame.

The NO concentration was quantitatively determined in the reference flame by measuring the LIF signal in the reference flame seeded with different known amounts of NO added in the premixed gases (from 0 to 600 ppm of NO was substituted for N_2 in the cold mixture). Measurements were performed at 20 mm above the burner. According to the modeling results, only 4% of the 600 ppm added in the premixed gases is removed due to the reburning process. The calibration method based on NO seeding was preferred to the one previously used in our lab using CRDS measurements [9]. Due to the line-of-sight absorption and the presence of background NO in the enclosure, CRDS measurements needed to be corrected according to a complicated procedure. The calibration of NO by LIF in seeded flames has been detailed in a few papers. Particularly, Thomsen et al. [16] insisted on the background effects on NO-LIF calibration. Berg et al. [17] studied the NO reburning process affecting the seeded NO in the calibration procedure, particularly under relatively high-seeding conditions (up to 3% NO added in the premixed gas) in lean and in rich methane flames. These difficulties have been thoroughly taken into account in this work. In the slightly lean reference flame, we noticed, according to [16], that both LIF signals (on and off resonance) increased linearly with the amount of added NO. Without NO added in the premixed gases, the off-resonance-corrected LIF signal ($\mathrm{LIF}_{(\mathrm{on-off})}$(native NO)) is proportional to the native NO. When NO was added in the premixed gases, the off-resonance-corrected LIF signal ($\mathrm{LIF}_{(\mathrm{on-off})}$(total NO)) is proportional to the total NO in the burned gas (native NO + added NO). Fig. 2 shows that the off-resonance-corrected LIF signal 20 mm above the burner depends linearly on the NO added in the reference flame indicating negligible reburning. The native NO was determined according to the inverse of the slope of the quantity (($\mathrm{LIF}_{(\mathrm{on-off})}$(total NO)/$\mathrm{LIF}_{(\mathrm{on-off})}$(native NO)) − 1) as a function of the added NO. In the reference flame, at 20 mm above the burner, the native NO was found to be 11.6 ppm with an accuracy of 15%.

Due to the very low LIF signal (especially in the flame containing VOC), the $Q_1(23)$ line was preferred to the $Q_2(27)$ one in the relative LIF profile measurements because it presents a rotational energy value (1910 cm^{-1}) lower than the one of the $Q_2(27)$ line (2322 cm^{-1}). But for the calibration measurements, the $Q_2(27)$ line was selected because the associated Boltzmann fraction is less dependent on the temperature than that of the $Q_1(23)$ line. In the reference flame, the relative NO LIF signal was high enough even when the $Q_2(27)$ line was selected. The comparison of the relative NO profiles measurements by exciting either the $Q_1(23)$ or the $Q_2(27)$ lines did not show any effect of the chosen excited line, except that the signal-to-noise ratio is much better in the former case.

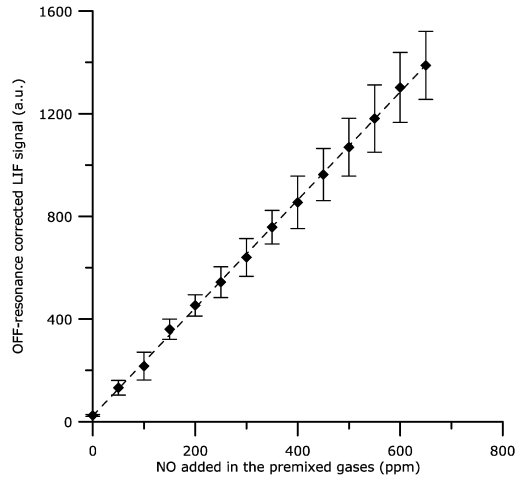

Fig. 2. Off-resonance-corrected LIF signal versus the known NO added in the reference flame at 20 mm above the burner. Vertical bars represent the uncertainties.

5. Modeling

5.1. Methodology

The premix code of the Chemkin-II package [18,19] was used during this work in order to perform modeling computations. Kinetic analysis is essentially based on the rate of production and consumption. The rate of production/consumption of each species is systematically computed using the appropriate subroutines in the Chemkin package (CKQYP, CKCONT) [19]. Following the Chemkin formalism, for each species k and each reaction i, the rate of progress is given by $R_{k,i} = \nu_{k,i} q_i$, where $\nu_{k,i}$ is the stoichiometric number relative to species k in reaction i, and q_i is the net rate of reaction i. The summation of $R_{k,i}$ over the I reactions is the production rate of species k. For species k, the involved reactions are sorted out in respect to their maximum absolute value. Their sign indicates if the species is consumed or produced. Then they are rearranged to determine the species flux when the same species are involved in both hands. The rate of progress depending on the species concentration and the temperature depends on the location as well. So, in order to better account for this variation along the burner axis, we choose to consider the species molar flux by integrating the rate of

progress versus the height above the burner. The integrated rate of progress for each reaction or group of reactions represents the contribution of this reaction or group of reactions to the species formation or consumption (according to the sign of the rate of progress).

All computations have been performed using experimental temperature profiles, which have been extended close to the burner surface at 400 K. The mass fluxes, computed with respect to the standard volumetric flow rate and the burner surface area, are reported in Table 1.

5.2. Initial mechanism

The detailed kinetic scheme devoted to NO formation prediction in methane/air flames containing VOC (MEK or EA) requires three submechanisms: (i) for the methane oxidation, (ii) for the VOC oxidation, and (iii) a full description of nitrogen chemistry.

The VOC submechanisms (for methyl ethyl ketone and ethyl acetate oxidation), proposed by Decottignies et al. [4] and Gasnot et al. [5], involve 156 reactions and 30 species. They both involve propanal, acetone, acetaldehyde, ethanol, DME (dimethyl ether), and methanol oxidation submechanisms. The MEK oxidation involve a methyl vinyl ketone (MVK) oxidation submechanism, while the EA oxidation involve acetic acid and vinyl acetate oxidation submechanisms. The VOC submechanisms were initially validated in conjunction with the GRI2.11 mechanism [20] for the methane oxidation. Full details concerning their validation were given in [4,5]. Hereafter this mechanism will be called the VOC-GRI2.11 mechanism.

The recent detailed mechanism GDF-Kin 3.0_NCN proposed by El-Bakali et al. [8] appeared to offer a good compromise for points (i) and (iii). It has been fully validated for natural gas oxidation in the presence of N species. It takes into account the new route of NCN contribution in the prompt NO formation. This model involves 883 reactions and 121 species.

In this work, the GRI2.11 mechanism was replaced by the GDF-Kin 3.0_NCN one. In the case of duplicate reactions (for example, as for the methanol or acetaldehyde oxidation submechanisms), the ones given in the GDF-Kin 3.0_NCN were adopted. The resulting mechanism, hereafter called the initial mechanism, has been associated with thermodynamic and transport databases initially considered. The reaction number affected to each reaction refers to the original order of the reactions of the GDF-Kin 3.0_NCN followed by the VOC submechanisms (starting from reaction 884).

5.3. Validation of the mechanism

The initial mechanism has been tested on the basis of the comparison between the experimental and the predicted profiles of CH radical and NO. Meanwhile, caution has been paid to preserve the initial overall good agreement between the experimental and the modeling profiles (according to the VOC-GRI2.11 mechanism) of the major species involved into the oxidation process of both methane and VOC. As a first step, the initial mechanism was tested according to the previous experimental data measured in low-pressure methane/O_2/N_2 flames containing different amounts of either MEK or EA [6,7].

The comparison between the experimental species profiles and those simulated showed that acetone was absolutely not predicted according to the initial mechanism. Decottignies et al. [4] showed that acetone is mainly produced by the recombination reaction between CH_3 and CH_3CO radicals (956) $CH_3COCH_3 = CH_3 + CH_3CO$. It appeared that the rate constant of the reaction (244) $CH_3CO + M = CH_3 + CO + M$, initially selected according to Wilk et al. [21] in GDF-Kin 3.0_NCN, led to a very fast consumption of the CH_3CO radical, inhibiting acetone formation. So we introduced the value, first selected in the VOC submechanisms [4,5] according to the recommendations of Baulch et al. [22] and recently confirmed by ab initio calculation [23]. Finally, in order to improve the acetone prediction by the model, the rate constant of the reaction (956) proposed by Sato and Hidaka [24] was selected.

Modification of the methane oxidation due to the replacement of the GRI2.11 mechanism by the GDF-Kin 3.0_NCN one affects principally the C_2H_6 and C_3H_8 simulated profiles. Based on the experimental profiles reported by Gasnot et al. [6,7] and the present ones (for CH and NO), few rate constant adjustments were proposed, as reported in Table 2. The involved reaction rate constants (except for reactions (244), (956), (967), and (985)) were initially estimated by analogy with the unsaturated hydrocarbon species according to Warnatz [25] for H-abstraction reactions and Dean [26] for thermal degradation reactions. Some oxygenated species profiles are given in Fig. 3; dimethyl ether or DME (CH_3OCH_3), acetone (CH_3COCH_3), methanol, methyl vinyl ketone or MVK ($C_2H_3COCH_3$), vinyl acetate (CH_3COO-C_2H_3), and acetaldehyde (CH_3HCO). The simulated profiles that are issued from the VOC-GRI2.11 mechanism and from the final version of the present work are compared to the experimental profiles in a stoichiometric CH_4/VOC/O_2/N_2 flame containing 2% in volume of VOC. Globally, the VOC-GRI2.11 and the new mechanism lead to similar predictions for the VOC intermediates profiles and therefore the final ki-

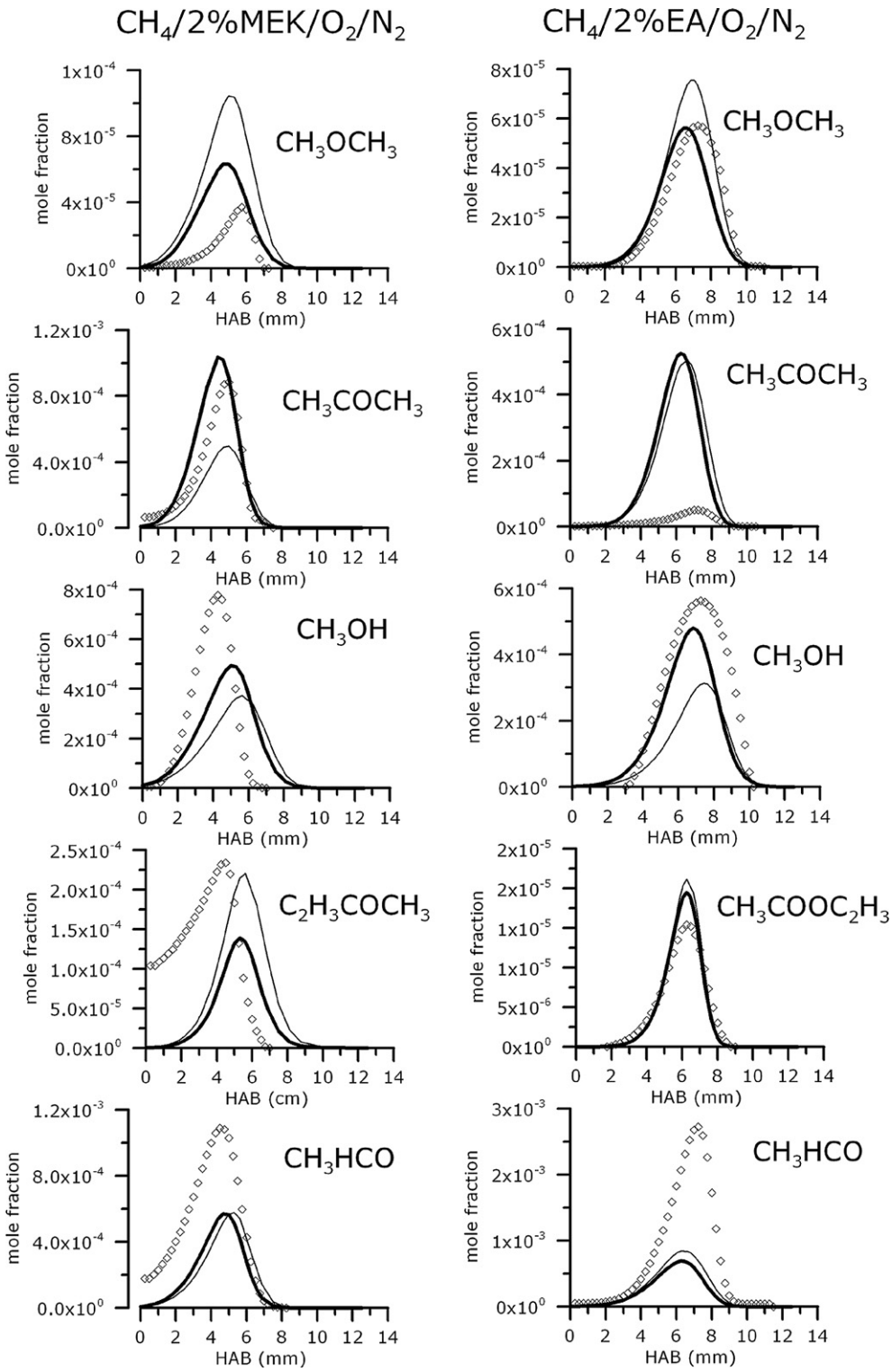

Fig. 3. Experimental oxygenated species profiles measured by probe sampling techniques in stoichiometric methane/VOC/O_2/N_2 flames containing 2% VOC [6,7]. Comparison with simulated data using previous kinetic mechanism (VOC-GRI2.11) (thin lines) and the present kinetic mechanism (thick lines); see text.

Table 2
List of reactions whose kinetic parameters have been adjusted compared with GDF-Kin 3.0_NCN (reaction 244) [8] and the initial VOCs submechanisms [4,5]

	Reaction	Rate constant	Adj.	Ref.
244	$CH_3CO + M = CH_3 + CO + M$	$k_\infty = 2.8 \times 10^{13} \exp(-17{,}260/RT)$; $k_0 = 8.428 \times 10^6$		[22]
910	$C_2H_5COCH_3 = CH_3 + C_2H_5CO$	$k = 2.0 \times 10^{16} \exp(-75{,}200/RT)$	/5	[26]
912	$C_2H_5COCH_3 = CH_3 + CH_3COCH_2$	$k = 5.0 \times 10^{17} \exp(-89{,}890/RT)$	∗5	[26]
913	$C_2H_5COCH_3 + H = CH_2CH_2COCH_3 + H_2$	$k = 4.69 \times 10^7 T^{2.0} \exp(-7703/RT)$	*5	[25]
914	$C_2H_5COCH_3 + H = CH_3CHCOCH_3 + H_2$	$k = 9.0 \times 10^5 T^{2.0} \exp(-5000/RT)$	/5	[25]
915	$C_2H_5COCH_3 + H = C_2H_5COCH_2 + H_2$	$k = 4.69 \times 10^7 T^{2.0} \exp(-7703/RT)$	∗5	[25]
920	$C_2H_5COCH_3 + OH = CH_3CHCOCH_3 + H_2O$	$k = 1.30 \times 10^8 T^{1.25} \exp(-703.4/RT)$	/5	[25]
935	$CH_3CHCOCH_3 = CH_2CHCOCH_3 + H$	$k = 4.5 \times 10^{16} \exp(-38{,}170/RT)$	∗5	[26]
956	$CH_3COCH_3 = CH_3 + CH_3CO$	$k = 5.65 \times 10^{16} \exp(-81{,}700/RT)$	∗1.9	[24]
967	$C_2H_5 + OH = C_2H_5OH$	$k = 4.345 \times 10^{14}$	∗5	[31]
985	$C_2H_5O + O_2 = CH_3HCO + HO_2$	$k = 2.4 \times 10^{15} \exp(-5017/RT)$	∗5	[32]
1000	$CH_3COOC_2H_5 + H = CH_3COOCH_2CH_2 + H_2$	$k = 2.25 \times 10^6 T^{2.0} \exp(-5000/RT)$	/2	[25]
1003	$CH_3COOC_2H_5 + O = CH_3COOCH_2CH_2 + OH$	$k = 1.3 \times 10^{13} \exp(-5215/RT)$	/2	[25]
1006	$CH_3COOC_2H_5 + OH = CH_3COOCH_2CH_2 + H_2O$	$k = 3.2 \times 10^8 T^{1.25} \exp(-703/RT)$	/2	[25]
1018	$CH_3COOCHCH_3 = CH_3CO + CH_3HCO$	$k = 1.0 \times 10^{12} \exp(9230/RT)$	/2	[26]
1019	$CH_3COOCHCH_3 = CH_3COOC_2H_3 + H$	$k = 1.2 \times 10^{13} \exp(-2708/RT)$	/2	[26]

Note. Numbers refer to the reaction number. Adj. indicates the adjustment factor to the preexponential constant compared to the rate constant value initially adopted in the VOCs submechanisms.

netic scheme will deliberately not be modified in the following. The discrepancy between the predicted and experimental profiles in the EA flame was already discussed in [5].

Fig. 4 shows the MEK rate of consumption as a function of the height above the burner. Computations were performed according to the VOC-GRI2.11 mechanism (Fig. 4a) and the present validated one (Fig. 4b). Adjustment of the rate constant values for the MEK consumption (implying reactions 910 to 920 presented in Table 3) involved an increase of the importance of the reactions 913 and 915 compared to the reaction 912.

Finally, the proposed detailed kinetic model is based on the kinetic mechanism GDF-Kin 3.0_NCN, in which the rate constant of the reaction (244) was updated and adjusted according to the recommendations of Baulch et al. [22] on one hand, and on the VOC submechanisms for MEK and EA oxidation proposed by Decottignies et al. [4] and Gasnot et al. [5], in which 15 rate constant values have been adjusted. The updated VOCs submechanisms are available by request from the authors.

5.4. VOCs oxidation schemes

In order to describe the VOCs' oxidation pathways, the relative importance of each reaction is determined according to the maximum value of its absolute net reaction rate. This determination has been found necessary to interpret the following discussion related to the CH and NO formation in seeded flames. The VOC's oxidation is governed by either H-abstraction or thermal degradation reactions which are not necessarily occurring at the same location above the burner. As reported in Fig. 4 in the case of the flame 2-MEK, the thermal degradation reactions (910 and 912) occurred 1 mm downstream from the H-abstraction reactions. So, in the present work, the relative pathways importance was established according to the molar species flux. Figs. 5 and 6 show the main pathways of the MEK and EA oxidation, respectively. The arrow thickness represents the relative pathway importance according to the ratio of the species molar flux of the involved reaction or group of reactions over the overall species molar flux. The underlined named species characterized the species that were previously measured by sampling techniques [4,5]. Numbers beside the arrows refer to the characteristic reaction numbers, whose rate constant has been adjusted. Those whose importance in the CH_3 formation has been highlighted, are underlined.

Considering the flame 2-MEK conditions as reported in Fig. 5, the main pathways of MEK degradation consist in H-abstraction reactions leading to $CH_2CH_2COCH_3$, $C_2H_5COCH_2$, and $CH_3CHCOCH_3$ radicals in the following percentages of MEK consumption: 34.8%, 34.8%, and 9.7%. MEK is thermally degraded, leading to CH_3COCH_2 and CH_3 radicals and to C_2H_5CO and CH_3 radicals, respectively, for 14.2 and 6.4% of the MEK removal.

Considering the flame 2-EA conditions as reported in Fig. 6, EA is consumed via H-abstraction reactions yielding $CH_2COOC_2H_5$, $CH_3COOCHCH_3$, and $CH_3COOCH_2CH_2$ radicals for 24.8, 12.1, and 7.3%, respectively. The thermal degradation leads to

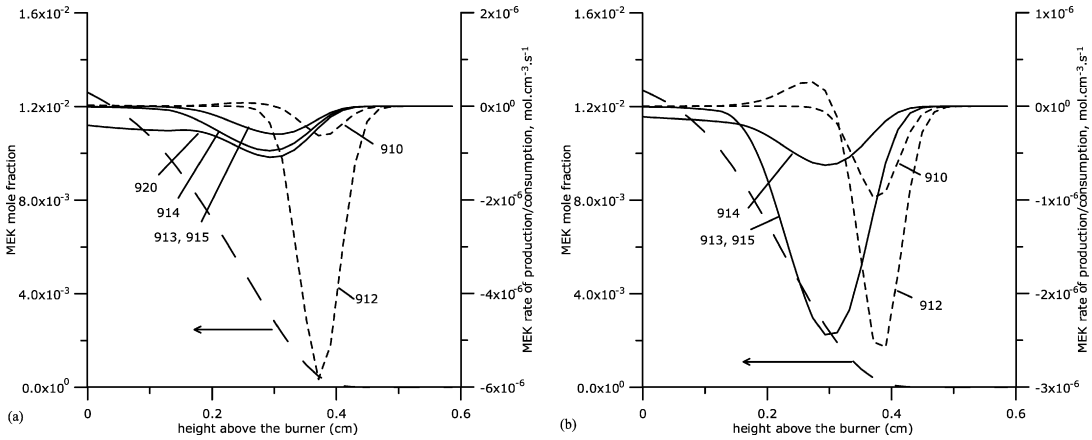

Fig. 4. MEK rate of consumption/production in the flame 2-MEK according to (a) the VOC-GRI2.11 mechanism and (b) the present model. Dotted lines represent thermal degradation reactions, solid lines represent H-abstraction reactions, and dashed lines represent MEK profile. Numbers besides the lines correspond to the reaction numbers listed in Table 2.

Table 3
Experimental CH-peak mole fractions (ppm) measured by LIF in different flames

Flame	CH peak (ppm)	CH peak rel	CH area rel	VOC/fuel (%)	C-atoms ($\times 10^{23}$)	CH_4 rel	Q_{fuel} rel	C-atoms rel	CH_3-VOC/CH_3 total (%)
Ref/MEK	1.69	1.00	1.00	0.0	5.59	1.00	1.00	1.00	0.0
1-MEK	1.55	0.92	0.99	5.4	6.11	0.89	0.94	1.09	8.0
2-MEK	1.26	0.75	0.78	21.8	6.84	0.58	0.74	1.22	24.7
3-MEK	1.05	0.62	0.60	50.9	7.54	0.26	0.53	1.35	46.4
Ref/EA	1.90	1.00	1.00	0.0	6.26	1.00	1.00	1.00	0.0
1-EA	1.90	1.00	0.86	10.2	7.35	0.81	0.90	1.17	13.2
2-EA	1.15	0.60	0.64	29.0	8.35	0.51	0.71	1.33	28.5

Note. The relative value of the experimental CH-peak and the determined CH-area are indicated. Considering the mixtures composition (see Table 1), the following values are reported: the VOC percentage in the fuel (VOC/fuel (%)), the C-atoms flow rate (atom/h), the relative values of CH_4, fuel volumetric flow rate and C-atoms. According to the integrated CH_3 rate of progress, the VOC contribution into the CH_3 formation is given (see text).

C_2H_5OCO radicals for 15.9% and acetic acid and ethylene for 39.7% of the EA removal. One can notice that the EA oxidation scheme is more dense than the MEK oxidation scheme.

6. Results and discussions concerning CH and NO profiles

6.1. Experimental profiles

CH and NO profiles, combined with the temperature profile, are plotted versus the height above the burner (Fig. 7). Temperature profiles measured by thermocouple are reported, showing that the burned gas temperature is close to 1800 K in all the studied flames. Experimental temperature profiles were extrapolated to zero position with a value of 400 K. NO is formed in the flame front area where the CH radical is present according to the prompt-NO mechanism. The NO profile is constant in the burned gases zone, indicating negligible contribution of the thermal-NO mechanism. Nevertheless, it is also noteworthy that the NO production starts upstream from the CH radical onset, as previously shown in several works [9,27]. According to kinetic considerations, NO is formed in the reaction zone while the experimental and simulated NO profiles show that NO is present upstream the reaction zone close to the burner surface. This might be due to preferential species diffusion towards the cold burner surface.

The absolute uncertainties during the calibration procedure have been shown to be 15% for CH and NO in the reference flame. However, the relative uncertainty in the LIF signal measurements during the linking process between the different flames was better than 10%, which is fairly good considering the very low CH peak and NO mole fraction (a few ppm). Therefore, any variation greater than 10% of the LIF signal between the investigated flames can be interpreted as being representative of an actual species variation in the flame. CH profiles measured in flames

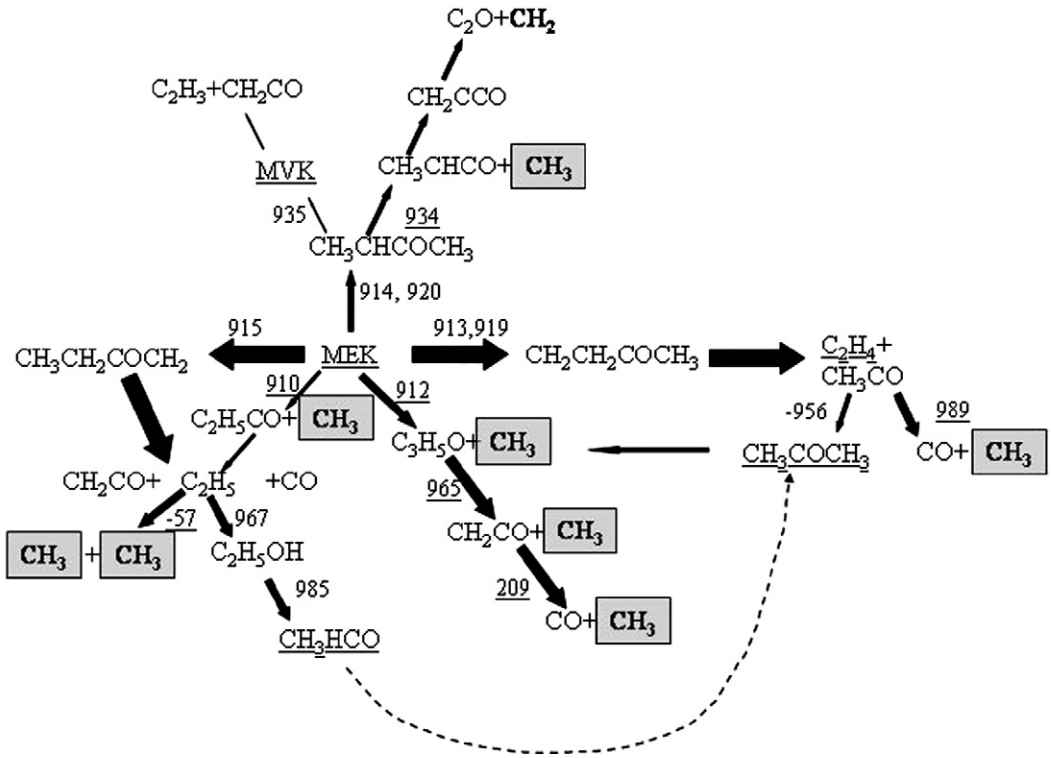

Fig. 5. MEK oxidation scheme. The arrow thickness depends on the contribution of the route to the species consumption in the flame 2-MEK. Underlined species were measured by GC sampling [4]. Numbers beside the arrows indicate the involved reaction number.

1-MEK and 1-EA are close to those measured in flames ref/MEK and ref/EA, respectively. The same observations can be done with NO profiles. The CH-peak values are reported in Table 3 for the different studied flames. It seems that the CH peak mole fraction decreases more rapidly in the presence of EA than in the presence of MEK. However, as the equivalence ratio is higher with the EA mixtures, this could probably affect the CH variation.

Both CH peak values and NO mole fractions in the burned gases were found to diminish when methane was replaced by VOCs. In Fig. 8, the NO mole fractions in the burned gases are plotted as a function of the CH peak mole fraction for numerous low-pressure flames in which the width of the CH profile is quite similar. Despite the different flame conditions (fuel, pressure, equivalence ratio, nitrogen dilution), it is noteworthy that NO and CH mole fractions depend on each other through a close to linear representation.

6.2. *Comparison between experimental and simulated profiles*

The CH radical is well known to be produced from the C1 sequence involving CH_3, CH_2, and CH radicals. Methane substitution by VOC in our flames leads to a CH_4 decrease in the premixed gases and could partially explain the observed CH radical diminution. However, the added VOC is expected to produce CH radicals also. Modeling study will make it possible to determine the exact contribution of VOC substitution to CH variation.

Experimental profiles have been compared to the simulated ones using the present model. In the three methane/O_2/N_2 flames, the agreement between experimental and simulated profiles is good. This result was expected due to the wide validation of GDF-Kin 3.0_NCN [8].

When methane is partially replaced by MEK or EA, the simulated CH peak values decrease in overall good agreement with the experimental results, as shown in Fig. 9, while the opposite effect was observed with the GRI2.11-VOC or the initial mechanisms.

Similarly, Fig. 10 presents a comparison between the experimental and simulated profiles of NO. The modeling reflects the VOC substitution effect on the NO formation well. According to the prompt-NO mechanism, the CH peak mole fraction decrease, previously observed with the VOC seeding, leads to a substantial NO decrease in the burned gases. The gradient of the NO concentration is particularly well pre-

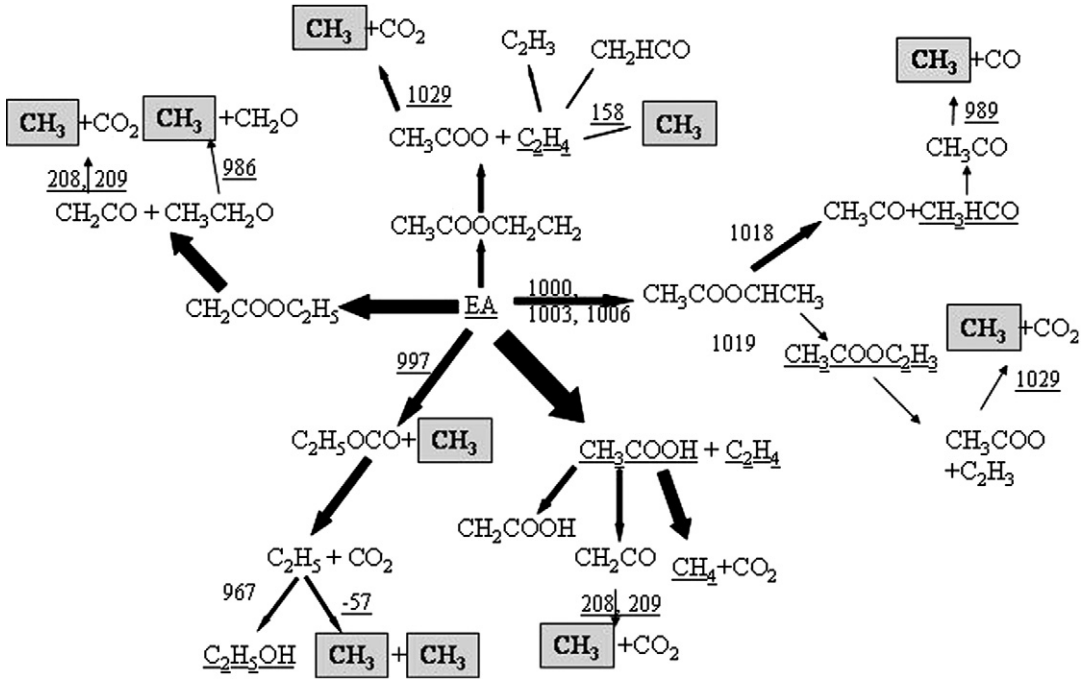

Fig. 6. EA oxidation scheme. The arrow thickness depends on the contribution of the route to the species consumption in the flame 2-EA. Underlined species were measured by GC sampling [5]. Numbers beside the arrows indicate the involved reaction number.

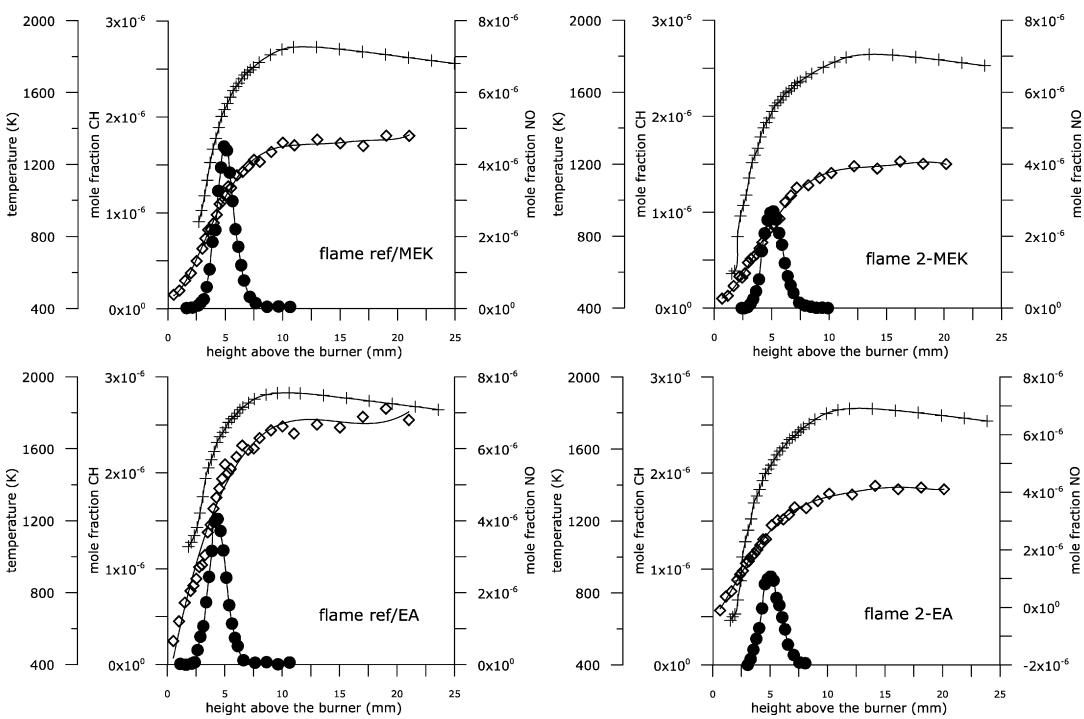

Fig. 7. Experimental profiles of (+) temperature, (●) CH, and (◇) NO in different flames stabilized at 40 Torr; see Table 1 for details.

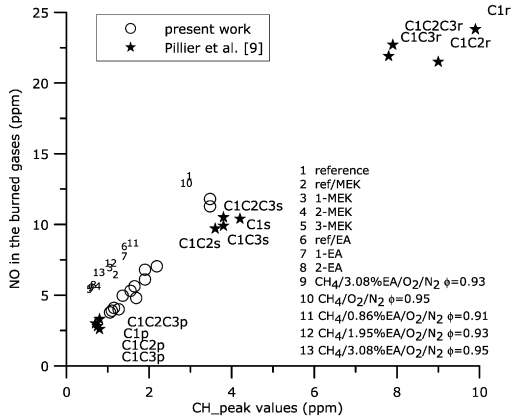

Fig. 8. NO density in the burned gas as a function of the CH peak values for different flames stabilized at 40 Torr. Results concerning additional flames (number 9–13) are reported. Experimental data (at $P = 33$ Torr) after Pillier et al. [9] are reported with respect to their flame labels; r: $\phi = 1.25$, s: $\phi = 1.0$, p: $\phi = 0.7$.

dicted in all the investigated flames. Predicted NO values in the burned gases are within 20% of the experimental ones.

6.3. Discussions

Considering the good agreement between experimental and simulated profiles, modeling has been used to analyze the CH formation pathways in order to understand the VOC impact on the CH formation and consequently on the (prompt) NO formation. CH radicals being mainly formed through the C1 sequence, the CH_3 pathways are first detailed and compared under four different flame conditions, ref/MEK and 2-MEK, and ref/EA and 2-EA, in order to highlight the VOC contribution to the CH formation via the CH_3 route.

6.3.1. CH_3 rates of production and consumption

The rates of production of the CH_3 radical were calculated according to the detailed kinetic model proposed in this work, under different flame conditions. Comparison between the pathways in seeded and nonseeded flames has made it possible to characterize a specific CH_3 route of formation when VOC is added in the gas.

In methane/O_2/N_2 flames, the analysis has shown that the CH_3 radical (well known to be formed by H-abstraction of CH_4 reactions) is mainly produced according to only three major steps: (39) $CH_4 + OH = CH_3 + H_2O$, (40) $CH_4 + O = CH_3 + OH$, and (41) $CH_4 + H = CH_3 + H_2$. According to the CH_3 molar flux, these three reactions are responsible for 95% of the total amount of CH_3 production.

When methane is replaced by MEK in the premixed gases, new steps are responsible for the CH_3 radical formation: (965) $CH_3COCH_2 = CH_2CO + CH_3$, (−57) $CH_3 + CH_3 = C_2H_5 + H$, (912) C_2H_5-$COCH_3 = CH_3 + CH_3COCH_2$, (209) $CH_2CO + H = CH_3 + CO$, (989) $CH_3CO + OH \Rightarrow CH_3 + CO + OH$, (910) $C_2H_5COCH_3 = CH_3 + C_2H_5CO$, and (934) $CH_3CHCOCH_3 = CH_3CHCO + CH_3$. These reactions are highlighted in the MEK oxidation scheme shown in Fig. 5. The reactions involved in the CH_3 formation have been divided into two groups; the first one characterizes the CH_4 oxidation contribution, while the second one represents the CH_3 specifically issued from MEK oxidation. Fig. 11a shows the CH_3 rate of production due to both fuels as a function of the height above the burner with respect to the

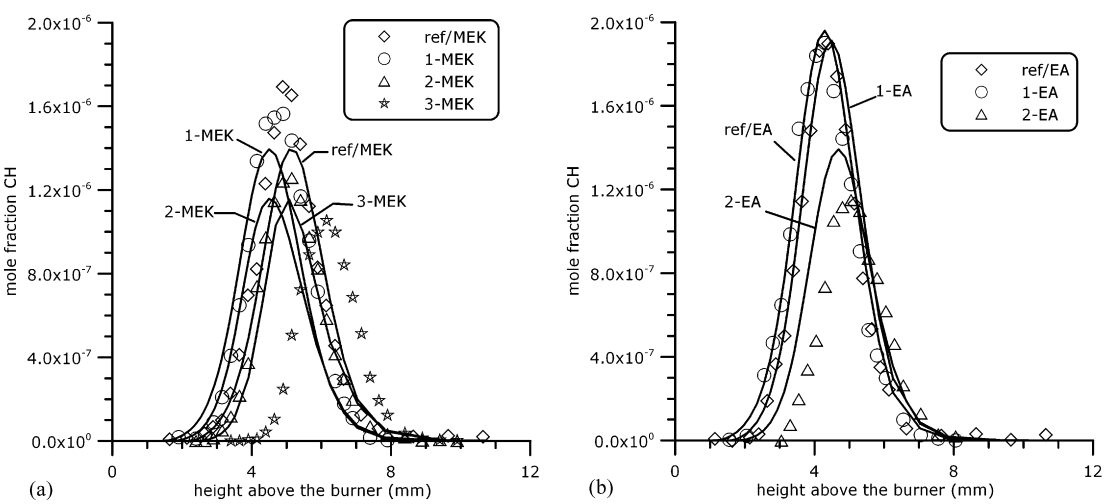

Fig. 9. Comparison between experimental (symbols) and simulated (lines) CH profiles (a) in flames containing MEK and (b) in flames containing EA. Flame compositions are given in Table 1.

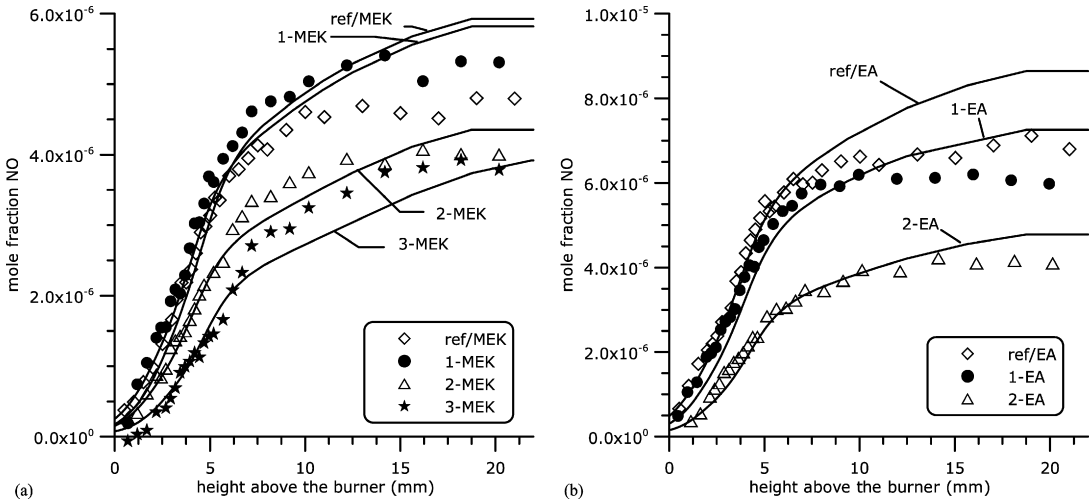

Fig. 10. NO profiles measured by LIF excitation (symbols) in the different flames for which the gas composition is reported in Table 1. Comparison with simulated profiles (lines) in the presence of (a) MEK, (b) EA in the flame.

flame 2-MEK condition. According to the CH_3 rate of production, 24.7% of the total amount of CH_3 is produced from MEK oxidation, while MEK represents 22% of the fuel in the mixture.

The same procedure has been performed with the EA-containing flames. The following ten reactions are specifically responsible for the CH_3 radical formation in the presence of EA in the mixture: (997) $CH_3COOC_2H_5 = CH_3 + C_2H_5OCO$, (−57) $CH_3 + CH_3 = C_2H_5 + H$, (986) $CH_3CH_2O = CH_3 + CH_2O$, (209) $CH_2CO + H = CH_3 + CO$, (965) $CH_3COCH_2 = CH_2CO + CH_3$, (158) $C_2H_4 + O = CH_3 + HCO$, (1029) $CH_3COO = CH_3 + CO_2$, (989) $CH_3CO + OH \Rightarrow CH_3 + CO + OH$, (245) $CH_2HCO = CH_3 + CO$, and (208) $CH_2CO + OH = CH_3 + CO_2$. These reactions are highlighted in the EA oxidation kinetic scheme shown in Fig. 6. As previously, both fuels' contributions to CH_3 formation were identified and their rates of production are plotted in Fig. 11b. In flame 2-EA, containing 29% of EA in the fuel, the respective integrated rates of production show that 28.5% of the CH_3 radical comes from the EA oxidation.

CH_3 radical rates of consumption have been examined and do not depend on the source of CH_3 formation. CH_3 consumption yields mainly CH_2O (∼50%), $^1CH_2$ (∼23%), CH_4 (∼8%), and CH_3OH (∼5%). In the presence of VOC, a very small part of CH_3 (<2%) leads to acetone (through reaction (956)). Simulated CH_3-peak values are plotted as function of the CH-peak values (Fig. 12) and show a strong correlation with the CH-peak values owing to the C1 sequence.

The VOC contribution to the CH_3 formation was computed in all studied flames, and the relative value is reported in Table 3 ((CH_3-VOC)/CH_3 total). One can notice that the percentage of CH_3 due to the VOC oxidation over the total CH_3 production is close to the percentage value of VOC in the fuel mixture, showing that one molecule of VOC presents the same ability for CH_3 formation as one molecule of methane. It must be emphasized, however, that a molecule of either MEK or EA contains four times as many carbon atoms as a molecule of methane.

6.3.2. CH radical formation

In order to take into account not only the CH-peak value, but the profile shape, one can determine the area of the CH profile. The values are reported in Table 3. In order to take into account the fuel amount variation in our flames and to make easier any comparison, relative values (labeled "rel") have been calculated according to the value obtained in the reference flames; ref/MEK for MEK seeded flames and ref/EA for EA seeded flames. One can notice that the relative CH-peak value is close to the relative CH-area value, indicating that the CH profile shape is similar from one flame to the other.

In our flame conditions, as reported in Table 3, the methane replacement by VOC implies a decrease of methane ($CH_{4\,rel}$) and also of the total fuel amount ($Q_{fuel\,rel}$) in the premixed gases, while the total number of C-atoms increases (C-atoms$_{rel}$). The relative behavior of CH-peak value in flames reflects the impact of methane decrease (which tends to reduce the CH radical production) and of the total number of C-atoms increase (which contributes partly to CH formation). It is noteworthy that the behavior of the relative CH-peak value is quite similar to that of the relative fuel volumetric flow rate. This is consis-

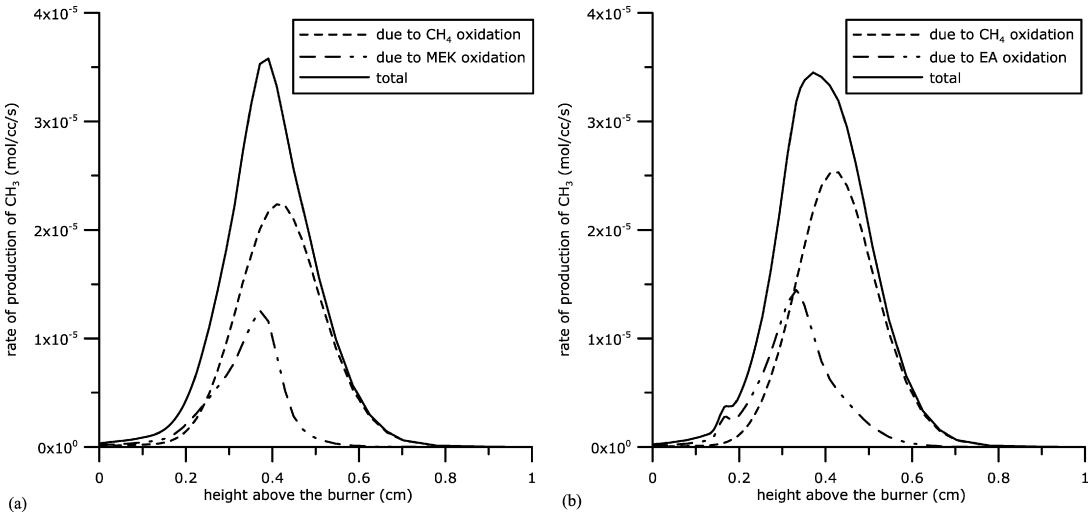

Fig. 11. Rate of production of the CH_3 radical in two different flames: (a) flame 2-MEK, (b) flame 2-EA. Contribution of CH_4 oxidation and VOC contribution to the CH_3 formation. The involved reactions are identified in the text.

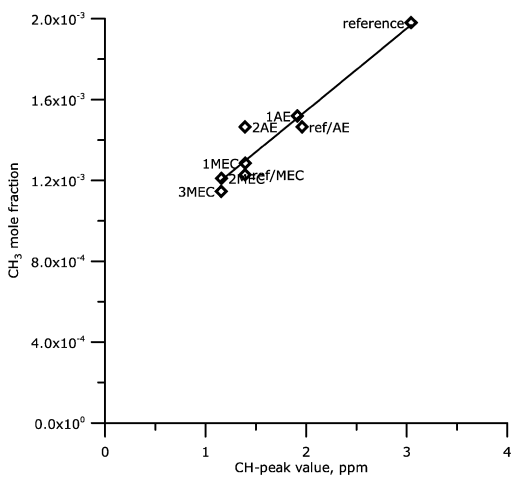

Fig. 12. Simulated CH_3-peak values vs the simulated CH-peak values according to the different flame conditions reported in Table 1.

tent with the fact that methane and VOC (MEK or EA) produce CH_3 radicals with a comparable efficiency (as reported in the previous section), which then is converted into CH radicals according to the C1 route.

With the two examined VOCs, when the oxygenated compound is added in the premixed gas keeping constant the equivalence ratio, the N_2/O_2 ratio, and the total flow rate, the total C-atoms increases but the CH-peak value decreases. This means that the oxygenated compounds are less efficient in the CH formation than CH_4. Consequently, NO formation mainly due to the prompt-NO mechanism decreases when methane is replaced by VOC.

6.3.3. NO formation

Recently, the NCN route in prompt-NO formation has been proposed as the new route instead of the classical HCN route [28,29]. A few authors have replaced the reaction $CH + N_2 = HCN + N$ by $CH + N_2 = NCN + H$ in kinetic models. El-Bakali et al. [8] found that the theoretical rate constant of the reaction $CH + N_2 = NCN + H$ proposed by Moskaleva and Lin [28] led to an important underestimation of prompt-NO formation. El-Bakali et al. [8] found that a rate constant ($k(\mathrm{cm}^3/\mathrm{mol\,s}) = 5.1 \times 10^{11} \exp(-13{,}600/RT)$) led to satisfactory agreement between experimental and simulated results. However, the kinetic rate constant of this reaction still requires validation. Full discussion of the rate constant value choice and validation in the GDF-Kin 3.0_NCN mechanism is proposed by El-Bakali et al. [8]. It has not been changed in the present version of the mechanism. The ability of this mechanism to predict NO formation has been confirmed in the present work in various lean VOC-containing flames exhibiting very low concentrations of CH and NO without any change of the nitrogen chemistry.

Williams and Fleming [30] recently proposed to include the C_2O route in the prompt-NO formation in order to take into account the stoichiometry and fuel type effect on the NO formation. So the possible role of C_2O as precursor has been considered here because of the propensity of VOC oxidation to form C_2O. As shown in Fig. 5, from $CH_3CHCOCH_3$ radical, CH_3CHCO is mainly formed with CH_3, leading to CH_2CCO and then to C_2O. Due to MEK, the maximum value of C_2O concentration can reach 100 times the value simulated without the MEK added, keeping constant the equivalence ratio. In the pres-

ence of EA in the premixed gas, the maximum value of C_2O density is close to the one simulated in the $CH_4/O_2/N_2$ flame, keeping constant the equivalence ratio and the N_2/O_2 ratio. If the C_2O route as NO precursor were considered in our final detailed kinetic mechanism, the NO formation would be widely affected. Therefore, in this work, the C_2O route does not appear satisfying for the NO formation in the investigated VOC-containing flames, and the NO route of formation was found satisfactorily described by the one initially postulated in the GDF-Kin 3.0_NCN mechanism.

7. Conclusions

LIF measurements of CH radicals and NO in low-pressure $CH_4/O_2/N_2$ flames containing either MEK (methyl ethyl ketone) or EA (ethyl acetate) have been performed in order to examine the VOC seeding effect on the NO formation. CH–LIF signal (using a predissociative state of CH: $C^2\Sigma^+ \leftarrow X^2\Pi(0,0)$ with a collection bandpass limited to the Q-branch) was calibrated using CRDS measurements. NO–LIF signal (using the $A^2\Sigma^+ \leftarrow X^2\Pi$ state by excitation of the (0–0) band and collection of the (0–2) band in order to minimize the background perturbation) was calibrated using NO seeded procedure. Special attention has been paid in order to take into account background effects [15,16] and possible NO reburning effects [17]. Relative NO–LIF signals were systematically determined from the difference between the on and the off resonance signals. During the calibration procedure, the same attention was followed by recording the LIF signals (on/off resonance) in the burned gases for different amounts of NO added in the flame.

Measurements along the flame have been realized for different flame compositions with a close to constant total volumetric flow rate. In a first series where MEK has been added to the mixture, the mixture equivalence ratio is close to 0.78. In a second series where EA has been added to the mixture, the equivalence ratio is close to 0.85. Temperature profiles have been measured using a coated Pt/Rh thermocouple.

A detailed mechanism for both MEK and EA oxidation has been validated according to CH and NO profiles. This mechanism consists of MEK and EA submechanisms previously proposed by Decottignies et al. [4] and Gasnot et al. [5]. These submechanisms initially incorporated with the GRI2.11 mechanism (for CH_4 oxidation) were validated under intermediates profiles measured using GC sampling techniques. In order to take into account the N_2 chemistry, the GRI2.11 mechanism [20] was replaced by the GDF-Kin 3.0_NCN one, recently proposed in our Lab [8]. Few rate constant values have been adjusted or updated in order firstly to maintain the initial good agreement between experimental and simulated profiles for the major and intermediate species, and secondly to improve the agreement between experimental and simulated species profiles of CH and NO.

When CH_4 is substituted by either MEK or EA in the flame, CH-peak values decrease similarly to the NO mole fraction in the burned gases, indicating that NO is mainly formed according to the prompt-NO mechanism. Considering the overall good agreement between experimental and simulated profiles, a modeling study has been performed in order to understand the VOC impact on CH radical and NO formation. According to the C1 sequence, analysis of the CH_3 radical formation, mainly based on species flux determination, made it possible to show that first the CH_3 radical is consumed independent of its source of formation; second one molecule of VOC (MEK or EA) has the same ability for CH_3 formation as one molecule of methane (even if the considered VOC contain four times as many C-atoms than CH_4). The CH-peak values decrease is close to the total fuel volumetric flow rates diminishing when methane is substituted by a VOC, which is consistent with the fact that a molecule of VOC presents the same propensity to the CH_3 formation as a molecule of CH_4.

Since the NO formation follows mainly the prompt-NO mechanism, the CH-peak values decrease implies that the NO mole fraction diminishes in the burned gases. So, the VOC oxidation leads to less NO release than a CH_4-based flame in respect to the equivalence ratio.

Acknowledgments

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